

II BE (Civil Engineering)

Class Test II October- 2023

SUB CODE: 3VLRG1

SUB. NAME: STRUCTURAL MECHANICS

Duration: 70 Min

Maximum Marks: 20

Note: Attempt any two questions. All questions carry equal marks.

- Q.1. A cast iron beam has a trapezium section as shown in figure-1. If tensile stress is not to exceed 100 N/mm^2 and compressive stress 80 N/mm^2 , what is maximum UDL the beam can carry safely over simply supported of span 4 m.

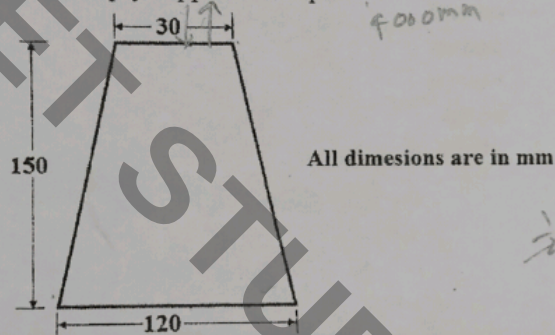


FIG - 1

Maxm ten $\sigma = 100 \text{ N/mm}^2$
 min $\sigma = 80 \text{ N/mm}^2$

$$S = \frac{I}{A}$$

$$= \frac{1}{12} \times 150 \times 150 \times \frac{120 + 30 + 120 \times 30}{3}$$

$$= \frac{1}{12} \times 150 \times 150 \times \frac{150}{3}$$

$$= \frac{1}{12} \times 150 \times 150 \times 50$$

$$= 1.277 \times 10^{-3}$$

- Q.2. Find the position and magnitude of maximum deflection for the beam as shown in figure-2. Take $E = 200 \text{ GPa}$ and $I = 7500 \text{ cm}^4$.

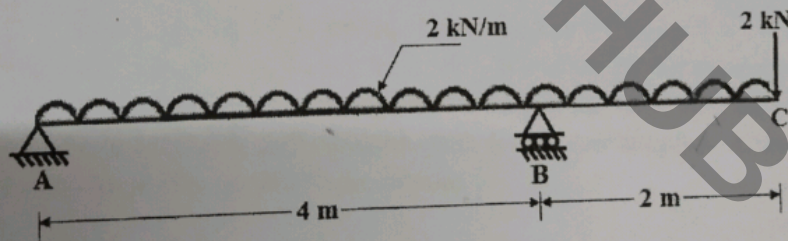


FIG - 2

- Q.3. Analyze the fixed beam as shown in figure-3 and draw its BMD.

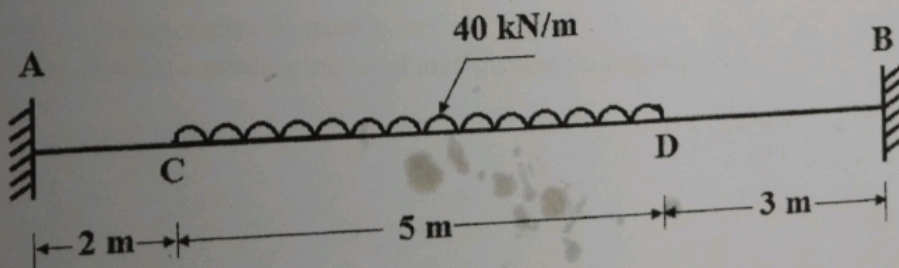


FIG - 3